Exp. 13

Pre-trained Model

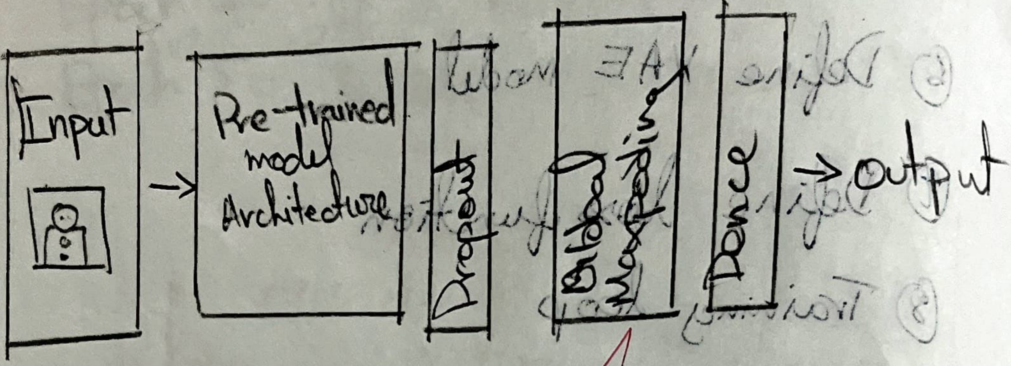
Aim:-

To understand the internal architecture and functioning of a pre-trained deep learning model for effective transfer learning and feature extraction

Objective:-

1. To explore the key components of pre-trained model such as convolutional, pooling, and fully connected layers
2. To analyze how pre-trained weights improve learning efficiency and accuracy
3. To extract and reuse learned features for a new classification task
4. To visualize the model's architecture and understand layer-wise data flow
5. To fine tune the pre-trained model to adapt to specific datasets or domains.

Pre-trained model Architecture



Pseudocode:-

BEGIN

1. Load necessary libraries
   Import deep learning framework

2. Load a pre-trained model

3. Explore the model architecture

4. Freeze base layers for feature extraction

5. Add custom classification layers

6. Train on target dataset

7. Evaluate and Visualize

END

Result

The code executed succesfully

Epoch	Loss
10	0.9452
20	0.7559
30	0.6089
40	0.4896
50	0.3957